

SEC P vs NP — notebook summary

SEC (autonomous) — attributed to Ludovico Kubler

2026-05-10 04:00 UTC

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Generated 2026-05-10 04:00 UTC

- Cycles recorded: **520**
- Time span: 390.3h (~1.33 cycles/h)
- Notebook: /home/ludo/Scrivania/SEC/research/pvsnp_notebook.jsonl

Verdict distribution

Verdict	Count
INCONCLUSIVE	485
BARRIER_HIT	16
FALSIFIED	15
SUPPORTED	4

Mathematical fields explored (field_A)

Field	Cycles
Real Algebraic Geometry	26
Free Probability	12
Schur-Weyl Duality	12
Representation Theory of Symmetric Groups	11
Matroid Theory	9
Algebraic Geometry	7
Random Matrix Theory	6
Additive Combinatorics	6
Finite Geometry	6
Spectral Graph Theory	6
Polymatroid Theory	6
Plethysm Theory	6
Ergodic Circuit Framework (communication complexity via dynamical systems)	4
{“framework_name”: “Ergodic Circuit Framework”, “math_branch”: “COMM_COMPLEXITY”}	4
Fourier Analysis on Boolean Functions	4
Algebraic Geometry of Secant Varieties	4
FOURIER_ANALYSIS	4
Fourier analysis of boolean functions	3
Algebraic Topology	3

Field	Cycles
matroid theory	3
Coarse Geometric Lifting (CGL) — Geometric Complexity Theory	3
{“framework_name”: “Tropical Circuit Weight Analysis (TCWA)”, “math_branch”: “BOUNDED_ARITHMETIC”}	3
Tropical Circuit Weight Analysis (BOUNDED_ARITHMETIC)	3
Schur-Weyl duality	3
Persistent Homology	3
Noncommutative Geometry	3
Noncommutative geometry	2
Representation theory of symmetric groups	2
Ergodic Circuit Framework (COMM_COMPLEXITY)	2
Tropical geometry	2

Last 15 cycles (chronological)

Time	Verdict	Title
2026-05-09 22:03 UTC	INCONCLUSIVE	p-adic Valuation of Solution Count Modulo p Inversely Proportiona
2026-05-09 22:15 UTC	INCONCLUSIVE	Slice Rank Lower Bound for Disjointness Communication Matrices
2026-05-09 22:28 UTC	INCONCLUSIVE	Symmetric Tensor Rank Gap in Permanent vs Determinant Decompositi
2026-05-09 22:50 UTC	INCONCLUSIVE	Standard Young Tableau Count Exponential Gap in Symmetric Power D
2026-05-09 23:33 UTC	INCONCLUSIVE	Plethysm Multiplicity Exponential Gap in Monotone Permanent vs De
2026-05-10 00:08 UTC	INCONCLUSIVE	Tropical Convex Hull Extreme Points Lower Bound for ACC ⁰ Circuit
2026-05-10 00:22 UTC	INCONCLUSIVE	Kronecker Coefficient Exponential Gap in Symmetric Powers of Perm
2026-05-10 00:30 UTC	INCONCLUSIVE	Operator Norm Lower Bound for Disjointness Communication Matrices
2026-05-10 00:53 UTC	INCONCLUSIVE	Tensor Rank Exponential Gap in Read-Twice vs Read-Once Branching
2026-05-10 01:02 UTC	INCONCLUSIVE	Projective Plane Line Count Bounds ABP Width for Monotone Circuit
2026-05-10 01:08 UTC	INCONCLUSIVE	Quantum Rank Lower Bounds for Disjointness Communication Matrices

Time	Verdict	Title
2026-05-10 01:23 UTC	INCONCLUSIVE	Symmetric Tensor Rank Gap in Permanent vs Determinant Decompositi
2026-05-10 01:47 UTC	INCONCLUSIVE	Negative Eigenvalue Count Lower Bounds SOS Degree for Max-CUT
2026-05-10 03:39 UTC	INCONCLUSIVE	Plethysm Coefficient Gap in Homogeneous Polynomial Decompositions
2026-05-10 03:58 UTC	INCONCLUSIVE	Hypergraph Maximum Matching Inverse Proportional to ACC ⁰ Circuit

How to read the reports

- reports/supported_findings.md — SUPPORTED conjectures with full test code. Paper-quality.
- reports/falsified_findings.md — FALSIFIED with counterexamples.
- reports/daily_YYYY-MM-DD.md — chronological log by day.
- pvsnp_notebook.jsonl — raw JSONL, one entry per cycle (source of truth).

PDF export

Install pandoc + a LaTeX engine once (sudo apt install pandoc texlive-xetex), then:

```
cd research/reports
for f in supported_findings.md falsified_findings.md notebook_summary.md; do
  pandoc "$f" -o "${f%.md}.pdf" --pdf-engine=xelatex -V geometry:margin=2cm
done
```